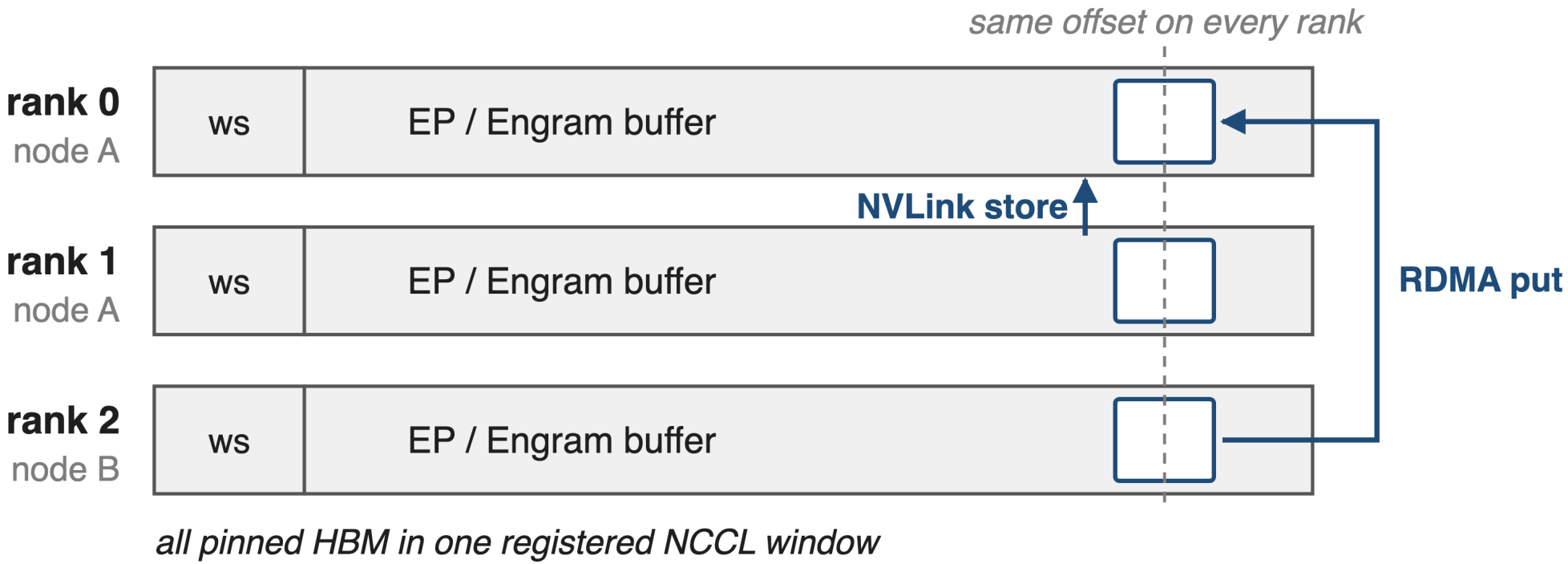
# Elastic GPU/CPU Hybrid Buffer

## One address range, HBM pages plus host pages

DeepSeek-AI · DeepEP V2 roadmap, shipped for Engram in V2.5

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

# DeepEP buffers are registered, symmetric ranges




*all pinned HBM in one registered NCCL window*

*A peer addresses a slot by (rank, offset); the size is fixed when the buffer is built*

# EP buffers reserve 8.9 GiB per GPU at EP64

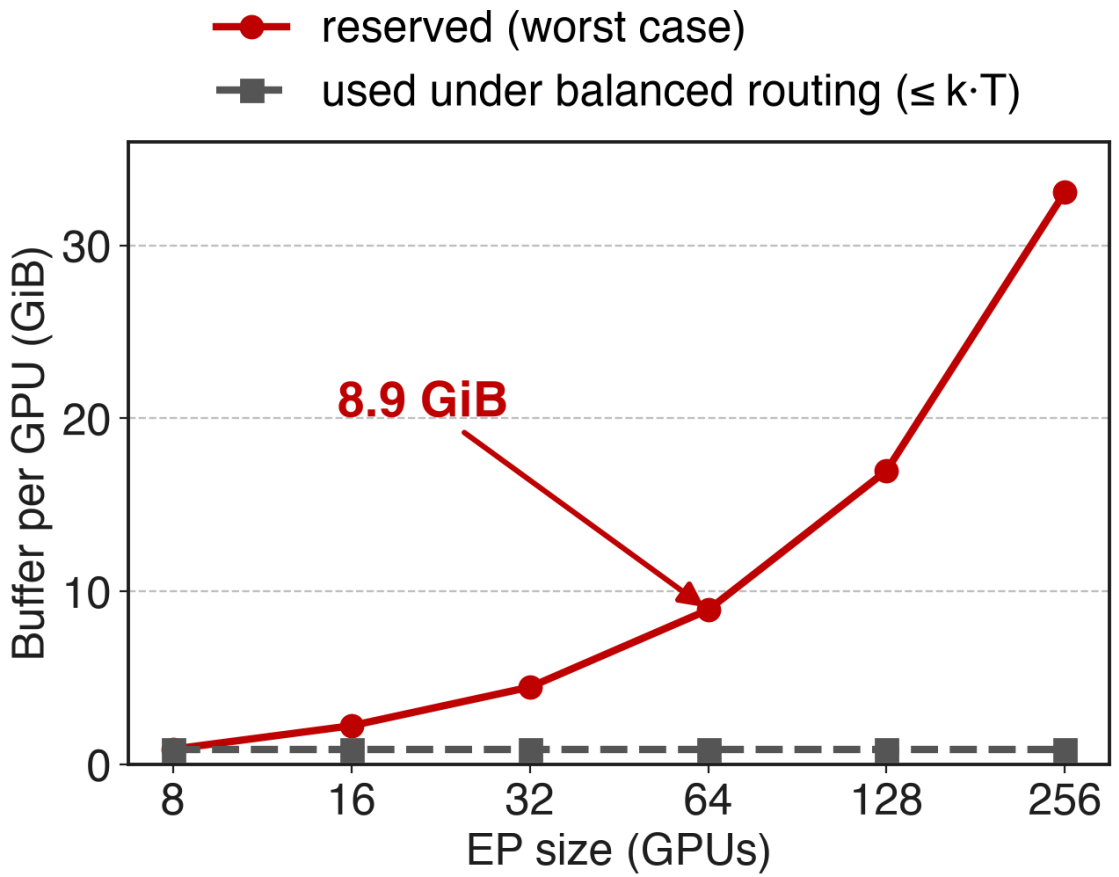
**8.9 GiB**

per GPU at EP64 with DeepSeek-V3-like settings: about 10× what balanced routing can use (analytic)

*Derived from DeepEP V2.5 EPBuffer size formula (csrc/buffers/ep.hpp) with the README benchmark settings*

# Worst-case sizing grows with EP size

- Any rank may receive **N·T** tokens
- Balanced routing delivers **≤ k·T**
- Gap grows linearly with EP size




*Analytic. T = 8,192 tokens/rank, hidden 7,168, top-8, FP8 dispatch, 8 GPUs per node*

# Engram tables outgrow HBM outright

- Engram-27B / 40B: **5.7B** / **18.5B** table parameters
- Offload test: **100B** parameters, **200 GB** in BF16
- One H100 holds **80 GB**

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**Each step reads only the activated rows, not the table**

# Today's options cost HBM or add a copy path

- **Keep it in HBM**
- Worst-case EP buffers, sharded tables
- **Capacity set by HBM**

- **Copy from host**
- Host prefetches rows over PCIe
- **Second address space, staging copies**

**Needed: host capacity behind the same registered range**

# Insight: an address need not imply HBM

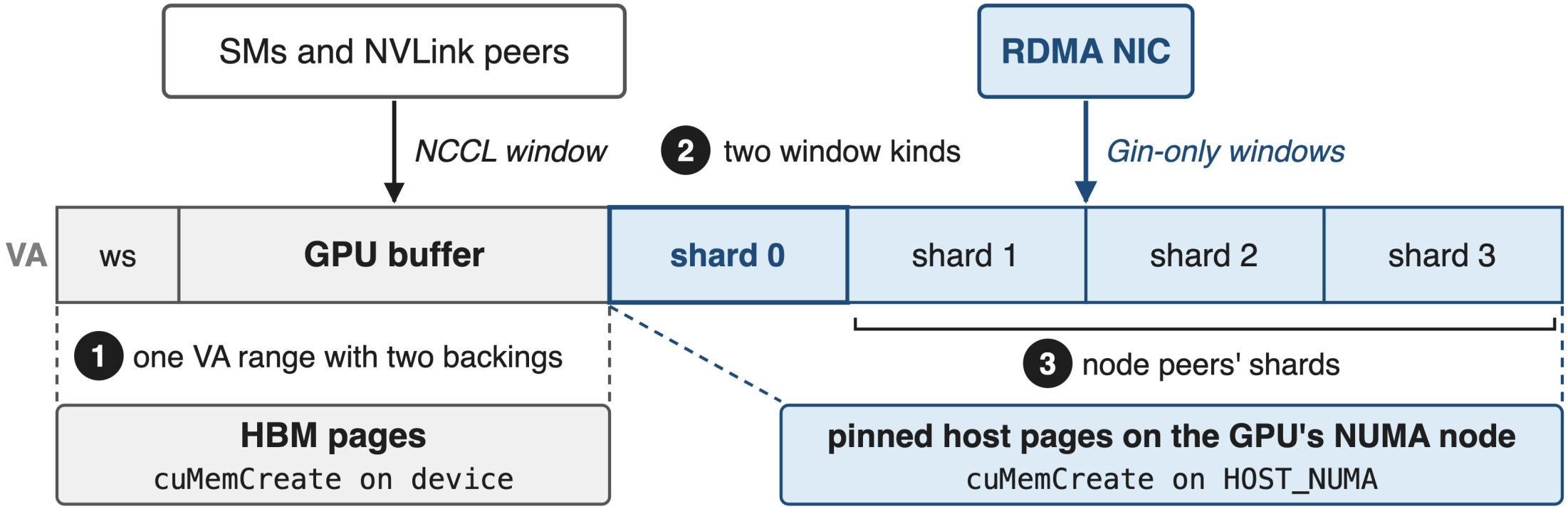
- CUDA VMM: reserve addresses first, back them later
- Backing: **HBM** or **pinned host pages**
- GPUDirect RDMA reaches both

# Insight: an address need not imply HBM

- CUDA VMM: reserve addresses first, back them later
- Backing: **HBM** or **pinned host pages**
- GPUDirect RDMA reaches both

**Keep one registered range; put only the hot bytes in HBM**

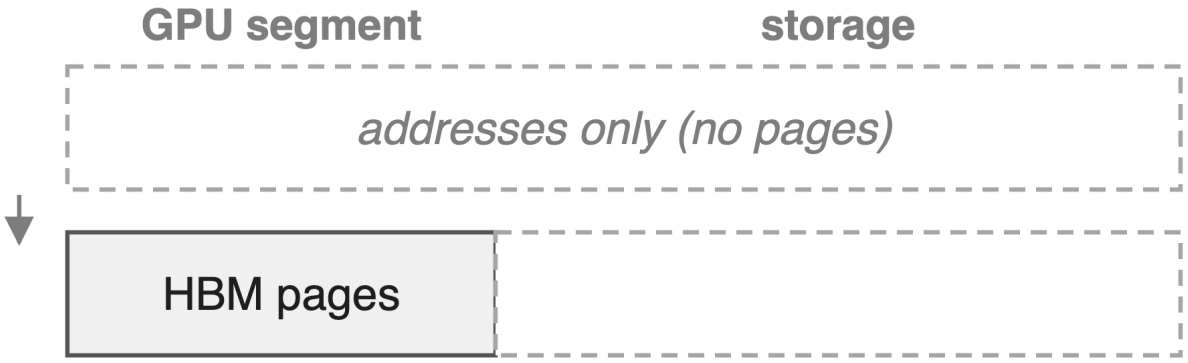
# Hot bytes in HBM, bulk in host pages




*DeepEP V2.5 symmetric.hpp: ElasticSymmetricMemory (one rank) and HybridElasticSymmetricMemory (node-local shards)*

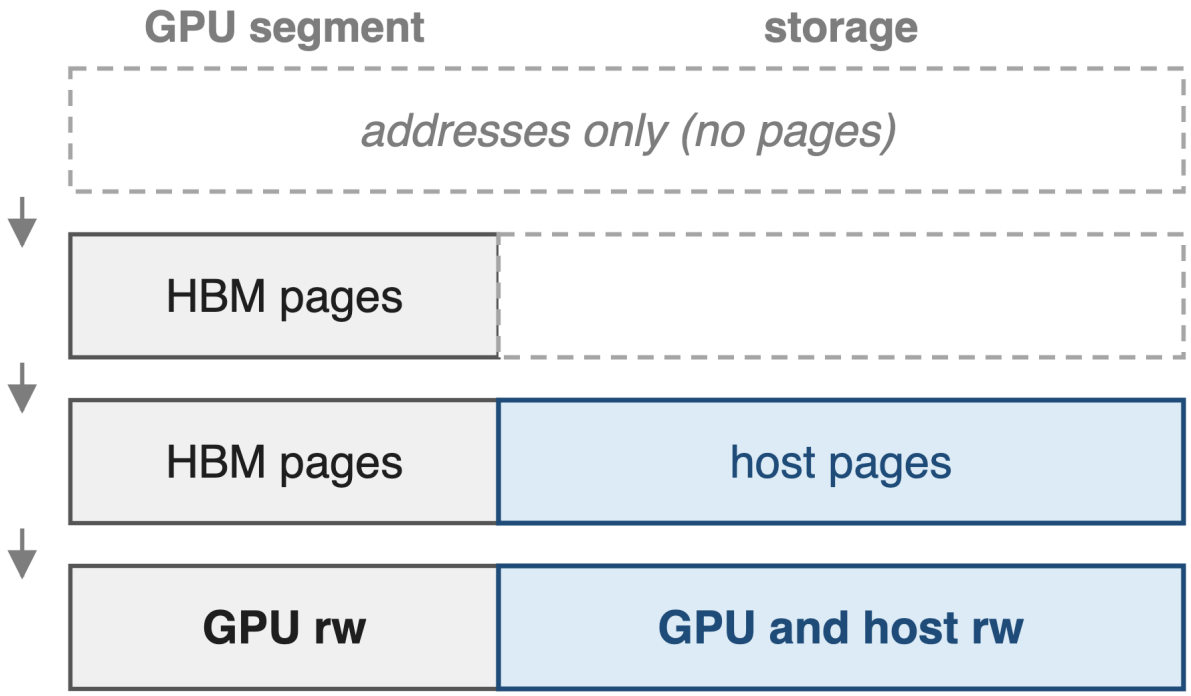
# ❶ Reserve once, map two kinds of pages

1. `cuMemAddressReserve`
   `(gpu + storage bytes)`
2. `cuMemCreate(DEVICE)`
   `cuMemMap(addr)`



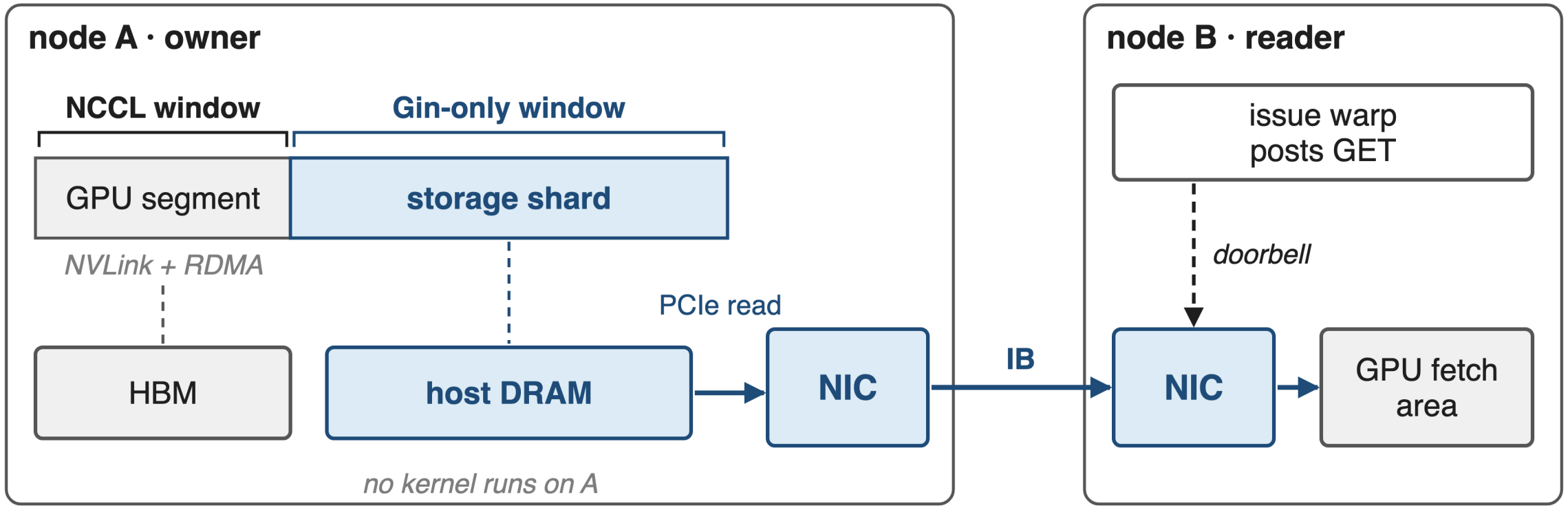
# ❶ Reserve once, map two kinds of pages

1. `cuMemAddressReserve`
   `(gpu + storage bytes)`
2. `cuMemCreate(DEVICE)`
   `cuMemMap(addr)`
3. `cuMemCreate(HOST_NUMA)`
   `cuMemMap(addr + gpu)`
4. `cuMemSetAccess`
   `(GPU + host NUMA)`



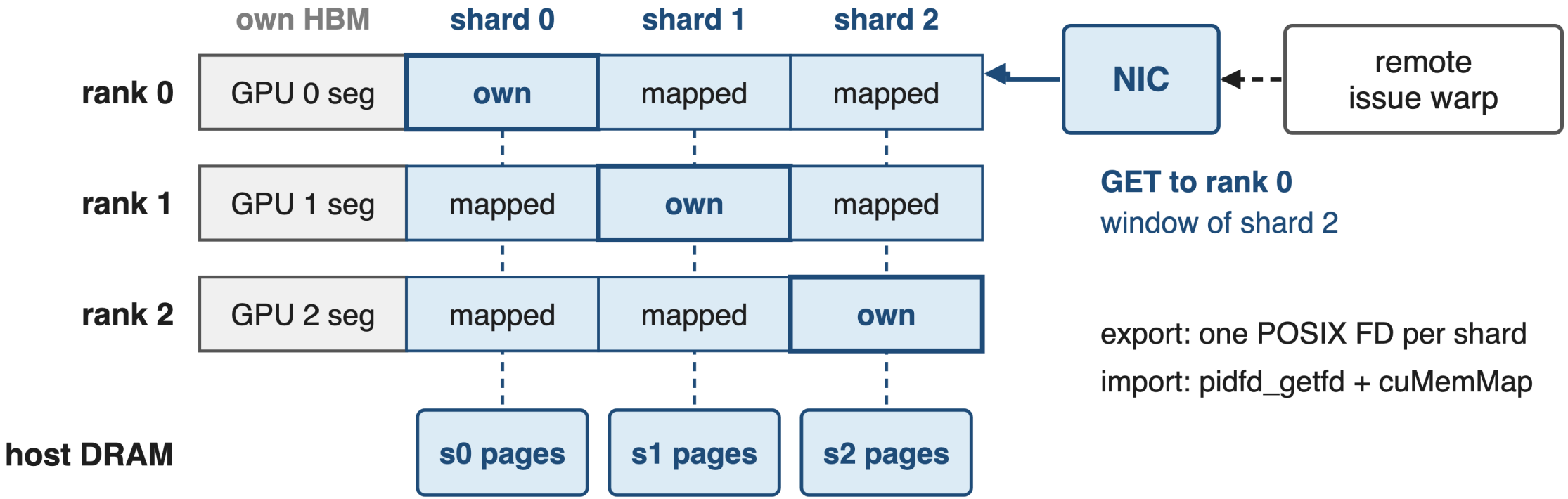

**One pointer and offset space; only the backing differs per range**

# ❷ Host shards are registered for RDMA only




**The owner runs no kernel; NICs move host rows into remote HBM**

# ❸ Every rank maps its node's shards




**One RDMA peer per node serves reads of every local shard**

# 12 MiB of HBM fronts 514 MiB of rows

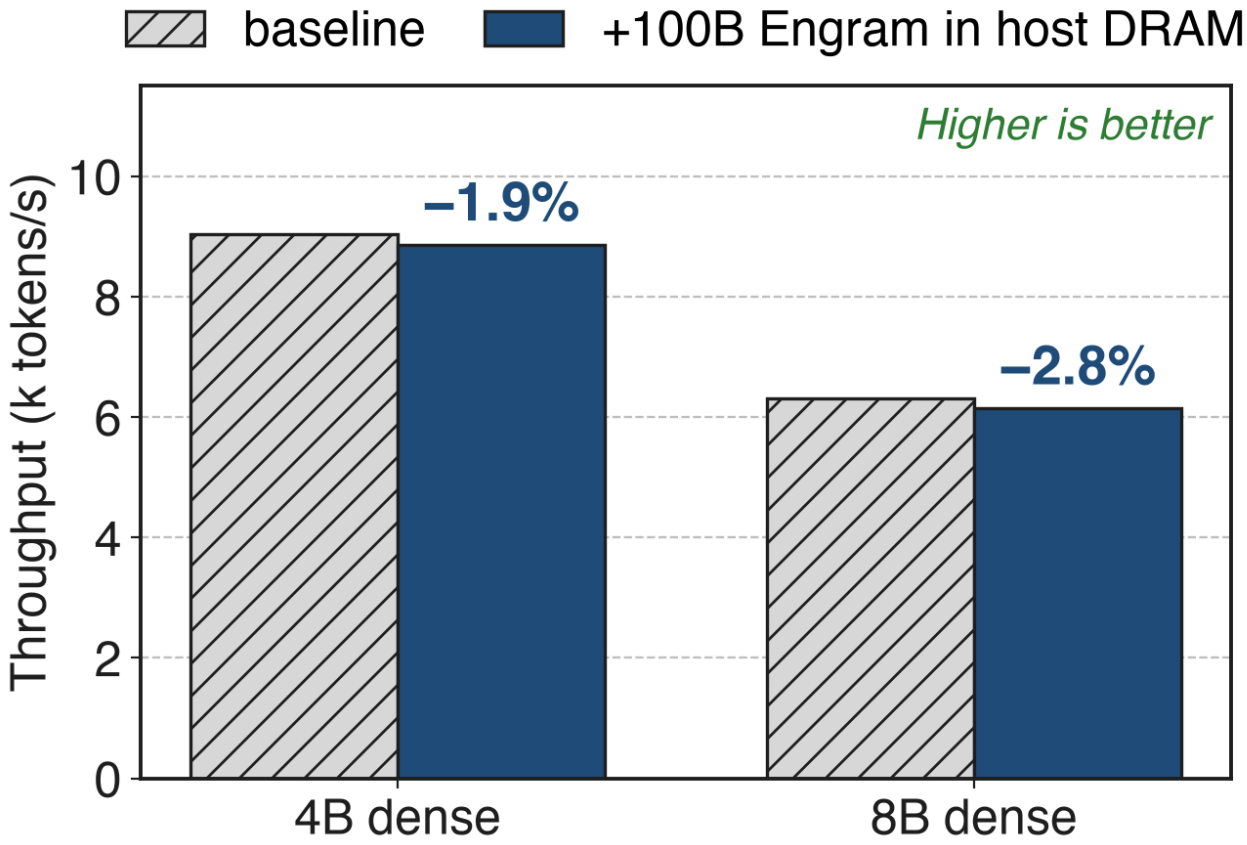
**12 MiB**

of HBM per rank serves 514 MiB of Engram rows kept in host DRAM: 43×
(DeepEP V2.5 test defaults, derived)

*EngramBuffer.get_storage_size_hint (csrc/buffers/engram.hpp) with tests/engram/test_engram.py defaults*

# A 100B table in host DRAM costs < 3%

- Engram paper, inference on H800
- Rows prefetched over PCIe, not DeepEP
- Indices known early hide the latency



# Engram ships it; EP overflow is roadmap

- **Shipped** (experimental)
- May 2026: first CPU bytes
- V2.5: `use_cpu_rdma_storage`

- **Roadmap** (V2 README)
- **EP buffers in host pages**
- **EP replay to shrink buffers**

**Needs GPUDirect RDMA on CUDA VMM; window stride set by env var**

# Summary

- EP buffers sized for the worst case: **8.9 GiB** per GPU at EP64, **~10×** balanced use (analytic)
- **Insight**: one VA range, HBM pages plus pinned host pages
- Engram: **12 MiB** of HBM fronts **514 MiB** of host rows per rank
- 100B table in host DRAM: **< 3%** throughput (Engram paper)

**https://github.com/deepseek-ai/DeepEP**